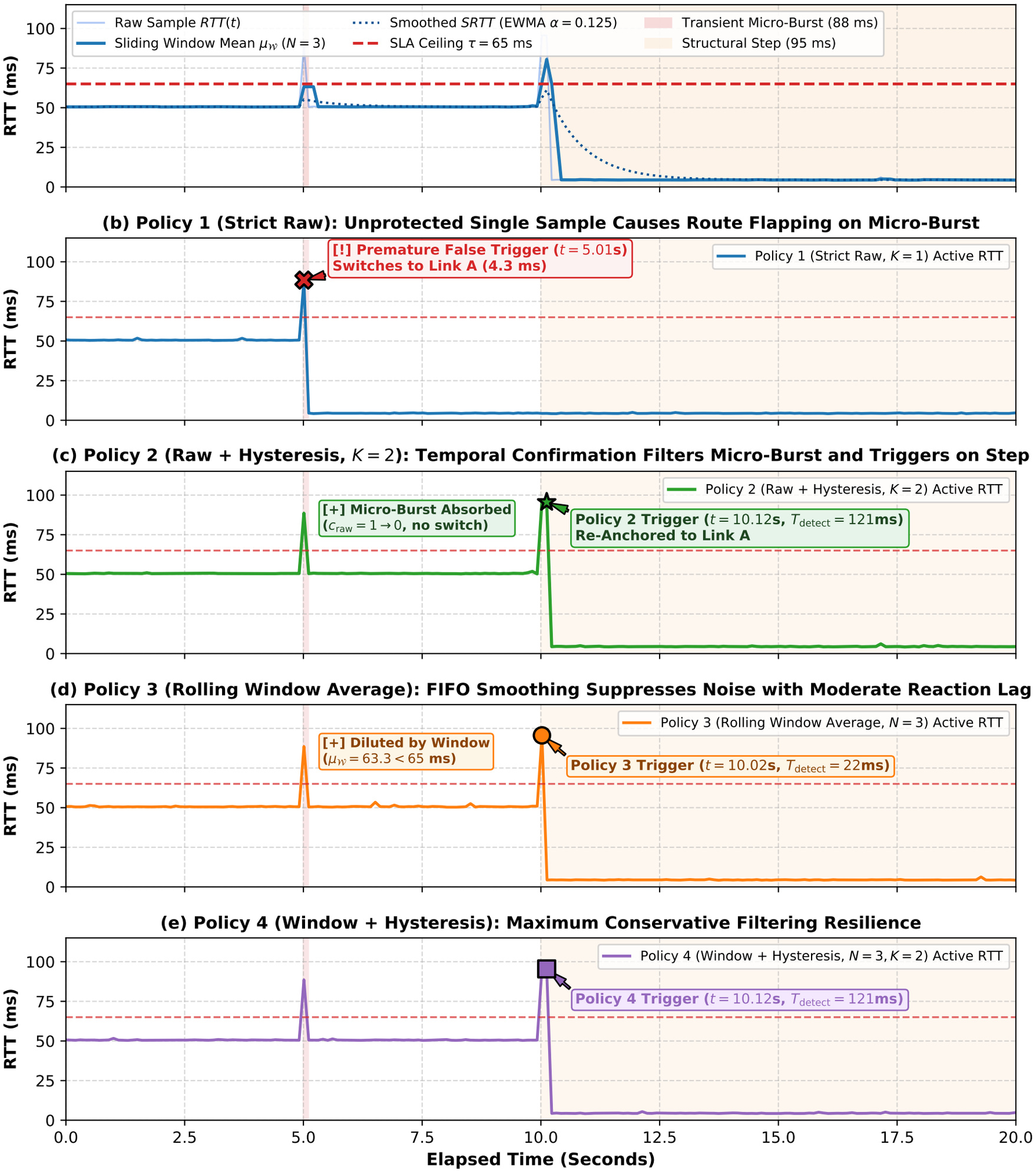

(a) Channel Scenario: 100ms Micro-Burst ($t = 5.0$s) and Persistent Degradation ($t = 10.0$s)
Raw Sample $RTT(t)$
Sliding Window Mean $\mu_{\mathcal{W}}$ ($N = 3$)
Smoothed $SRTT$ (EWMA $\alpha = 0.125$)
SLA Ceiling $\tau = 65$ ms
Transient Micro-Burst (88 ms)
Structural Step (95 ms)
RTT (ms)
(b) Policy 1 (Strict Raw): Unprotected Single Sample Causes Route Flapping on Micro-Burst
[!] Premature False Trigger ($t = 5.01$s)
Switches to Link A (4.3 ms)
Policy 1 (Strict Raw, $K = 1$) Active RTT
(c) Policy 2 (Raw + Hysteresis, $K = 2$): Temporal Confirmation Filters Micro-Burst and Triggers on Step
[+] Micro-Burst Absorbed
($c_{\text{raw}} = 1 \rightarrow 0$, no switch)
Policy 2 Trigger ($t = 10.12$s, $T_{\text{detect}} = 121$ms)
Re-Anchored to Link A
Policy 2 (Raw + Hysteresis, $K = 2$) Active RTT
(d) Policy 3 (Rolling Window Average): FIFO Smoothing Suppresses Noise with Moderate Reaction Lag
[+] Diluted by Window
($\mu_{\mathcal{W}} = 63.3 < 65$ ms)
Policy 3 Trigger ($t = 10.02$s, $T_{\text{detect}} = 22$ms)
Policy 3 (Rolling Window Average, $N = 3$) Active RTT
(e) Policy 4 (Window + Hysteresis): Maximum Conservative Filtering Resilience
Policy 4 Trigger ($t = 10.12$s, $T_{\text{detect}} = 121$ms)
Policy 4 (Window + Hysteresis, $N = 3, K = 2$) Active RTT
0
25
50
75
100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Elapsed Time (Seconds)